

SUPPLYGUARD AI

AI-Powered Supply Chain Control Tower

Supply Chain Disruption Assistant & Fleet Utilisation Optimizer

Team Phoenix

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Detect disruption → understand impact → optimize response → act

The Problem

Logistics teams need to make high-impact decisions while the network is changing.

DISRUPTIONS

Port strikes, severe weather, route closures and carrier capacity loss can cascade across the network.

BLIND SPOTS

Shipment risk, fleet availability and cold-chain telemetry often live in separate operational views.

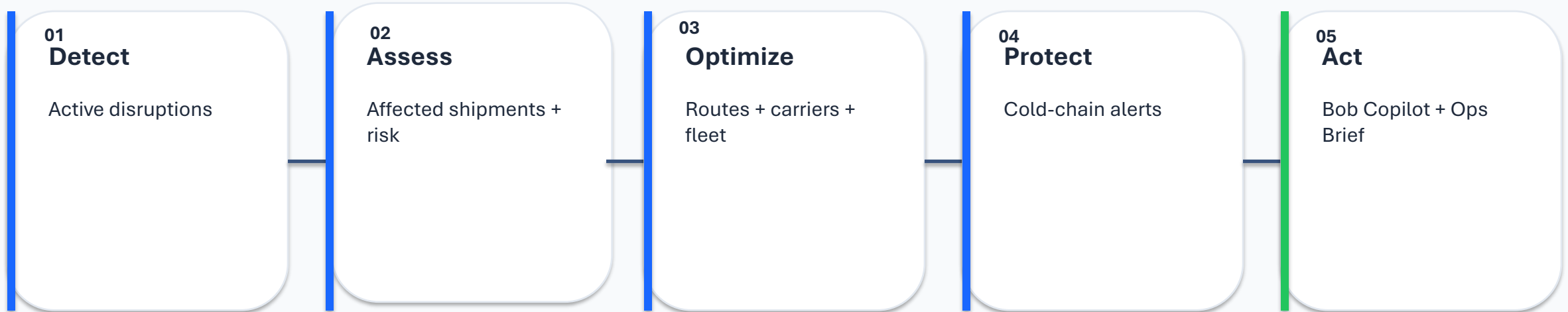
SLOW RESPONSE

Operators must manually identify affected shipments, compare alternatives and decide what to do first.

The result: delayed decisions, idle assets, missed deadlines and preventable cargo risk.

Our Solution

One operational intelligence layer connecting disruption, shipment, fleet and cold-chain data.

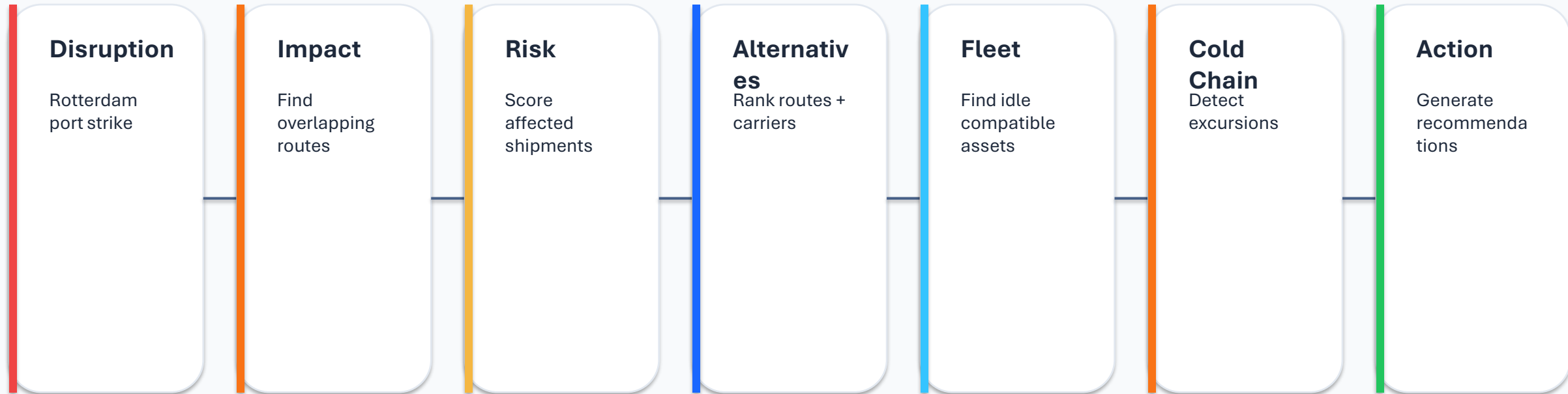


SupplyGuard turns fragmented signals into a prioritized action plan.

Decision support, not just monitoring

How the System Works

A disruption triggers a chain of analysis and recommendations.



Example: Rotterdam strike → affected shipment → CRITICAL risk → Antwerp route → idle truck → reduced delay.

Disruption Intelligence

The screenshot's Disruptions module is the entry point to the response workflow.

ACTIVE DISRUPTION: ROTTERDAM PORT WORKERS STRIKE

Severity: CRITICAL

Status: ACTIVE

Routes affected: 6

Estimated delay: ~168h

SupplyGuard traces the disruption into the logistics network instead of stopping at a static alert.

WHAT HAPPENS NEXT

1. Identify overlapping routes
2. Find affected shipments
3. Calculate delay + deadline exposure
4. Estimate cargo value at risk
5. Recommend response options
6. Surface priority actions to the operator

11 Active

2 Critical

Shipment Risk & Route Optimization

Every recommendation is tied to measurable operational factors.

SHIPMENT RISK

Risk score: 0–100

LOW: 0–24

MEDIUM: 25–49

HIGH: 50–74

CRITICAL: 75–100

Factors: disruption impact, delay, deadline, cargo value, priority and cold-chain exposure.

ROUTE OPTIONS

Compare alternatives using:

- travel time
- cost
- disruption exposure
- capacity
- reliability
- deadline impact

Return a ranked recommendation with an explanation.

CARRIER OPTIONS

Rank carriers by:

- available capacity
- reliability
- cost
- route compatibility
- disruption exposure

Show the best alternative and why it wins.

Fleet Utilisation Optimizer

Turn idle capacity into an actionable recovery option.

IDLE ASSET

TRK-204

Status: IDLE

Location: Antwerp

Capacity: 18 tons

Idle duration: 9 hours

MATCH

Compatible affected
shipment

SHP-1042

Potential benefit:
~14h delay reduction

REDEPLOY

SupplyGuard evaluates location +
capacity + shipment need.

REDEPLOY → assignment
changes in application state.

This is an operational action, not
a decorative button.

Goal: increase utilisation while reducing disruption impact.

Cold-Chain Intelligence

Protect temperature-sensitive cargo before delivery.

IOT TELEMETRY

Continuous temperature readings

Normal example:

5.1°C → 5.4°C → 6.0°C

Excursion example:

8.5°C → 9.2°C → 9.8°C

ANALYSIS

Detect:

- above-range excursions
- below-range excursions
- duration
- severity
- rate of change

Severity is driven by configurable demo rules.

ACTION

NORMAL → Monitor

WARNING → Monitor closely

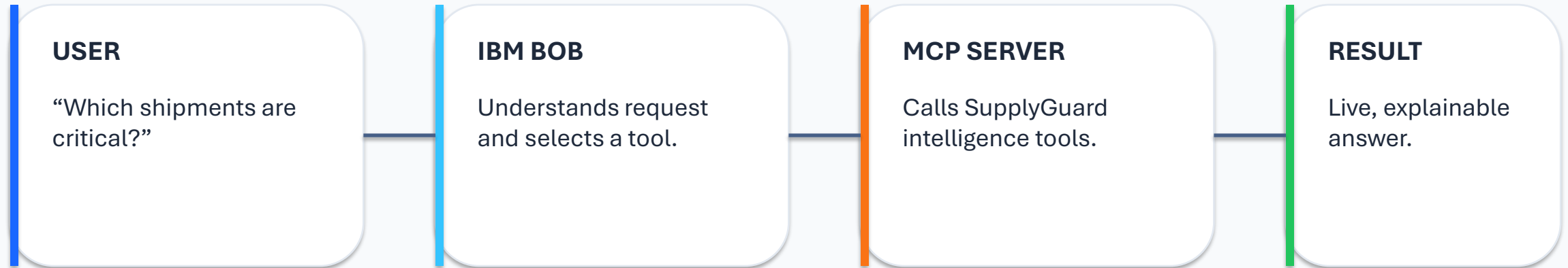
EXCURSION → Contact carrier / inspect

CRITICAL → Initiate quality review

Rules must be validated for production use.

IBM Bob Copilot + MCP

Natural-language access to real SupplyGuard intelligence.



REAL INTEGRATION

MCP tools:

get_active_disruptions • get_affected_shipments • analyze_shipment_risk
recommend_routes • recommend_carriers • get_idle_fleet
recommend_fleet_redeployment • get_cold_chain_alerts
analyze_temperature_excursion • generate_operations_brief

Example: "Give me the top five actions I should take right now."

Technical Architecture

Modular architecture keeps business logic testable and the demo

FRONTEND

React • TypeScript • Vite • Tailwind • Recharts

BACKEND

Node.js • Express • TypeScript • REST APIs

DATA

Synthetic shipments • disruptions • fleet • carriers • routes • IoT

INTELLIGENCE

Risk • Disruption • Route • Carrier • Fleet • Cold Chain

MCP / BOB

MCP server exposes intelligence tools to IBM Bob

All core decisions can run locally with synthetic data; external AI services are not a single point of failure.

Live Demo Story

A five-minute narrative that demonstrates the complete L2

ACTIVATE

Rotterdam Port Strike

IMPACT

Open affected shipments

DECIDE

Inspect critical shipment +
route/carrier alternatives

OPTIMIZE

Redeploy an idle fleet asset

PROTECT

Show cold-chain excursion +
severity

ASK BOB

“What should I do right
now?”

SUMMARIZE

Open Operations Brief

Impact & Differentiation

From visibility to action.

VISIBILITY

One control tower for

Disruptions

Shipments

Fleet

Cold Chain

Operations

INTELLIGENCE

Explainable risk scoring

Ranked alternatives

Asset-to-shipment matching

Temperature severity
analysis

ACTION

Reroute

Switch carrier

Redeploy fleet

Inspect cargo

Ask Bob

Prioritize next steps

Our strongest idea: connect every operational signal to a recommended next action.

What We Built & What Comes Next

A hackathon MVP designed to evolve into a production control tower.

TODAY — MVP

- ✓ Disruption impact analysis
- ✓ Shipment risk scoring
- ✓ Route + carrier recommendations
- ✓ Idle fleet redeployment
- ✓ Cold-chain excursion detection
- ✓ Bob Copilot via MCP
- ✓ Operations Brief
- ✓ Scenario-based demo

NEXT

- Connect live TMS/WMS data
- Stream real IoT telemetry
- Add predictive ETA / risk models
- Add role-based access
- Expand optimization constraints
- Production-grade governance & monitoring

BUILD • INNOVATE • ACT